Arnab Dey

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Professional Summary

Cloud and Platform Engineer with **15+ years in IT** and **9+ years focused on cloud and platform engineering** across Azure, AWS, OpenStack, and hybrid environments. I design and build with code — Terraform for infrastructure, Python and Bash for automation, GitOps for delivery — and favour reviewable, repeatable patterns over manual operations. Hands-on across the CAF Enterprise Scale / landing-zone pattern, Kubernetes (Rancher, K3s, EKS), GitHub Actions, Azure DevOps, Jenkins, ArgoCD, Prometheus/Grafana, and Azure Policy / RBAC. Consistent track record of moving platform teams off manual ClickOps toward self-service, audited, cost-aware infrastructure.

Core Skills

Infrastructure as Code — Terraform (AzureRM, AWS, OpenStack, Databricks providers), reusable module libraries, ARM templates, Ansible, archetype-driven CAF Enterprise Scale

Cloud Platforms — Microsoft Azure (primary), AWS (EC2, S3, VPC, EKS, IAM, CloudWatch), OpenStack (PAN-NET / private cloud), hybrid and on-prem

Scripting — Python (automation, CLI tooling, REST API integration), Bash, PowerShell

CI/CD & GitOps — GitHub Actions, Azure DevOps Pipelines, GitLab CI, Jenkins, ArgoCD, Git / Gerrit, trunk-based and GitFlow workflows

Containers & Orchestration — Kubernetes (Rancher RKE, K3s, EKS, kubeadm), Docker, Helm, Kubeflow

Identity, Security & Governance — Entra ID, Azure RBAC (including custom role definitions), Azure Policy, IAM, SAML federation, Key Vault, least-privilege patterns, backup-integrity by RBAC

Observability — Prometheus, Grafana, Azure Monitor, Log Analytics / KQL, ELK Stack, CloudWatch, cert-manager, sealed-secrets

Ways of Working — Agile / SAFe, cross-team platform enablement, technical documentation, stakeholder workshops, mentoring

Languages — English (fluent), Swedish (SFI C-level)

Professional Experience

DevOps Engineer — AFRY Digital Solutions AB

May 2025 – Present | Stockholm, Sweden

Assignment: Ericsson AB — CI Engineer, Tiger Test Tools (Sep 2025 – Present)

- Maintain and support the end-to-end CI flow for radio-communication smoke and regression testing — Git, Gerrit, Jenkins, Docker, Linux.
- Support radio-communication hardware and test servers; troubleshoot issues across daily delivery and validation workflows.

Assignment: AFRY Internal — True Demand Initiative, RPA Developer (Jul 2025 – Sep 2025)

- Built a hybrid Power Automate (Desktop + cloud) solution that processes Tendsign procurement emails end-to-end, extracts document links, and automates browser-based login and download with secure credential handling.

Platform engineering (ongoing, internal project)

- Designed and deployed a **GitOps-based single-node Kubernetes platform** (K3s + ArgoCD + Prometheus + Grafana + cert-manager + sealed-secrets) as a reference for cloud-native delivery patterns — maintained as a live public demo.

Cloud Engineer — Stena Metall AB

Aug 2024 – Jan 2025 | Gothenburg, Sweden

Cloud Infrastructure Engineer on the central Cloud Office (Governance side) of a mature **Azure CAF Enterprise Scale** estate — ~100 subscriptions, multi-management-group hierarchy, production workloads for a multi-billion-SEK industrial group.

- **Authored two custom Azure RBAC role definitions** (a "Backup Remover" role and a modified Contributor with explicit `notActions`) to enforce separation-of-duties between resource management and backup deletion — a ransomware-resilience control missing from Azure's built-in Contributor role. Wired into the CAF archetype framework and deployed via the existing Terraform + Azure DevOps pipeline. **38 commits across the core-governance repo.**
- Delivered a **tenant-wide Key Vault compliance audit** — KQL against Azure Resource Graph for soft-delete retention, purge protection, and RBAC authorisation — producing CSV inventories and a documented review for architecture leadership.
- Built **Service Principal lifecycle tooling** — KQL audit for 30-day create/update/delete events, PowerShell for expiring-credential inventory, and a design for client-secret rotation via Logic Apps + Graph API `addPassword`.
- Implemented **Azure Backup policy compliance reporting** with per-VM exempt-rationale tracking.
- Contributed PowerShell tooling: subscription budget automation, ACR token expiry inventory, .NET runtime enumeration across App Services, Hybrid Benefit enablement.
- Delivered fixes across multiple application landing-zone repositories (IdentityNow VA, Key Vault with disk-encryption-set, cost-allocation tagging, AzureRM version alignment).
- KQL-based monitoring contributed to a **~20% reduction in downtime** across supported services.
- Published sanitized sister repos on GitHub: `iac-azure-core-governance` (CAF Enterprise Scale governance platform), reusable Terraform module library, and Azure cost-optimisation templates.

Senior Software Engineer (DevOps) — Capgemini Sverige AB

Sep 2021 – Jul 2024 | Gothenburg, Sweden

Assignment: Volvo Cars — Car Safety R&D (Abakus MLOps / AI ART)

Self-service Databricks platform via GitOps

- Designed and owned a **GitOps self-service pattern for Azure Databricks**: data scientists request compute via a GitHub issue template → a YAML manifest PR → GitHub Actions runs `terraform fmt/validate/plan` for review → merge triggers `terraform apply`.
- Authored the **Terraform module** with opinionated, governance-friendly defaults baked in: single-user security mode, 30-minute auto-termination, autoscale 1→3 workers, cluster-log shipping to DBFS, `CAN_RESTART` permissions — making clusters cheap and safe by construction.
- Remote state in Azure Blob Storage with backend config injected at init from GitHub environment secrets; two-workflow contract (`tf-validate` on PR, `tf-deploy` on merge).
- **Outcome:** provisioning lead time dropped from days to same-day; every cluster traceable to a PR, a manifest, and a named owner; eliminated long-tail idle-cluster spend.

CAE HPC platform modernization (on-prem)

- Contributed to modernising a ~36-node CAE/HPC cluster supporting ML research: OS migration from SLES to Ubuntu LTS; stand-up of **Rancher-managed Kubernetes (RKE)** with worker fleet; **Kubeflow** installation via Helm for ML engineers.
- Built a **production-grade ELK stack** (Elasticsearch, Logstash, Kibana) on LXD with X-Pack security, SSL, and snapshot repositories for observability.
- Authored Ansible roles within the shared GitLab-managed codebase.

Dual-cloud ML storage benchmarking

- Built **two matched Terraform stacks** (AWS: EC2 + S3 + KMS + IAM | Azure: Linux VM + Premium BlockBlobStorage) as identically-shaped, disposable test rigs so data scientists could run MLCommons-style object-storage benchmarks on equal footing. Owned Terraform, IAM, dataset storage, and the tear-down pattern — the infrastructure was the control variable.

Other

- Led the internal **AI-powered Resume Builder Agent** side project (sanitized public repo).
- Maintained a Python automation toolkit for infrastructure and operational tasks (public repo).

Senior Business Consultant (DevOps) — Tech Mahindra

Feb 2017 – Sep 2021

Assignment: Cisco Systems — CD 2.0 on AWS (Oct 2019 – Mar 2020, and ongoing platform support)

- **Authored the Terraform** for Cisco's CD 2.0 continuous-delivery platform on AWS — VPC, subnets, security groups, EKS cluster, IAM roles, Jenkins on EKS, ECR integration, build-agent fleet, ALB, DNS — extending an internal reference architecture originally built on on-prem OpenShift.

- Wrote a separate **Terraform stack for an AWS observability environment** — EKS, Prometheus Operator via Helm, Grafana, persistent storage, ingress.
- **Closed out Cisco's long-running SVN-to-Git migration** by writing a **Python automation** that enumerated SVN repositories, extracted history, provisioned Bitbucket repos via REST API, pushed converted Git history, and produced CSV migration reports — with retry logic and batching to work around SCM rate limits.
- Onboarded application teams to CD 2.0 via CI engagement templates, Bitbucket/Crucible provisioning, and Jenkinsfile adaptation.

Assignment: Deutsche Telekom — 5G BSS Slicing Pilot on OpenStack (Apr 2020 – Jul 2021)

- Owned the **DevOps platform side** of DT's 5G Network Slicing commercial pilot on **PAN-NET (OpenStack)**, delivering Blue Marble's ~30-microservice BSS suite plus the Phenom UI.
- Wrote **Terraform against the OpenStack provider** — networks, subnets, security groups, VMs, floating IPs — provisioning the Jenkins / Nexus / Tuleap / Rancher / Keycloak / ELK estate.
- **Ansible playbooks** for all service configuration; custom Dockerfile for Sonatype Nexus 3 with SSL.
- Designed and ran **~46 Jenkins jobs** building BMP microservices, tagging with build number + short git SHA, pushing to internal Nexus registry, updating Kubernetes manifests, and triggering Rancher rolling updates.
- Owned the **cross-tenant migration** when the pilot moved PAN-NET tenants: wrote Python/shell automation to clone 54 Tuleap repositories at correct branches, re-seed Jenkins jobs, and rehome Nexus artifacts, Keycloak realm, and running Kubernetes workloads — cutover completed over a weekend, nothing rebuilt from scratch.
- Built a **Logstash pipeline** ingesting 5G UDR (Usage Data Record) CSV feeds into Elasticsearch for the billing emulator.
- Published a sanitized OpenStack DevOps solution on GitHub.

Assignment: Greenstone Financial Services — Insurance Portal (Sep 2019 – Jan 2020)

- Led DevOps transformation for .NET microservices: Azure DevOps pipelines, **Terraform for Azure infrastructure**, Docker + Docker Swarm with Azure Container Registry, Azure Monitor for observability.

Assignment: CA ANZ

- Architected the GitLab-based CI and source-control model — defined branching strategy, repository standards, pipeline conventions; supported team adoption through documentation and knowledge transfer.

Senior AEM Administrator (Cloud) — ICF Next (formerly CITYTECH)

Sep 2015 – Jan 2017 | Gothenburg / Remote

Assignment: Channel 7 (Australia) — AEM on AWS

- Built and operated a production Adobe Experience Manager platform on AWS for a national Australian broadcaster — VPC design, EC2 lifecycle (m3 family), Apache HTTPD + AEM Dispatcher hardening, SSL termination, and **SAML 2.0 federation** (Azure AD / ADFS → AWS IAM) replacing long-lived IAM user keys.

- Evolved the architecture from Phase-1 (separate Dispatcher VM) to Phase-2 (Dispatcher co-located on Gold Publisher; Publish tier in an Auto Scaling Group).
- Authored **Ansible roles** for AEM install, Apache/Dispatcher deployment, SSL, and system baseline — making instance rebuilds a runbook rather than a discovery exercise.
- Maintained the package-parity procedure between staging and production, catching release drift before deployments.

Unix / Linux Systems Administrator — HP Enterprise Services

Nov 2011 – Aug 2015 | India

Three-and-a-half years on HP Enterprise Services' Global Operations Center (Unix Service Delivery Unit) under the Best Shore Services model, rotating across three large outsourcing customers:

- **Hindustan Unilever** (~18 months, L1/L2) — 24×7 shift-based incident handling on a SAP-heavy Linux/Unix estate (140+ servers).
- **E.ON** (~12 months, L2/L3) — multi-site German/UK operations; storage, DNS, console management; started authoring SOPs.
- **Con-way Inc.** (~13 months, L2/L3 SOP author) — **authored a substantial portion of the BSS-ITO Con-way Linux procedure library** covering LVM filesystem resizes, Veritas SAN migration, kernel-parameter tuning, and ITO/OpsBridge agent management. Procedures used as team reference material.
- Tooling fluency: HP Service Manager, HP OpenView / OpsBridge, HP Server Automation (Opsware), NIM, HPSA.

Education

B.Tech in Computer Science — West Bengal University of Technology, Kolkata, India (2005 – 2009)

Certifications

- **Microsoft Certified: Azure Fundamentals (AZ-900)** — Mar 2023 *(Credential ID: 801D970BAA49297)*
- **Microsoft Certified: Azure AI Fundamentals (AI-900)** — *in progress*